STATS 101B - Week 4 Discussion

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Problem 1

The UCLA Stats Dept wishes to evaluate the impact of Supaclass HW grading on the final exam scores of STATS 101C. It believes that final exam scores may be influenced by both the presence of Supaclass grading and the professor. With all treatment combinations being possible, students are randomly assigned to each combination.

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- **Supaclass**: Yes, No
- **Class Sizes**: 80, 80, 80

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Problem 1 - Solution for Part A

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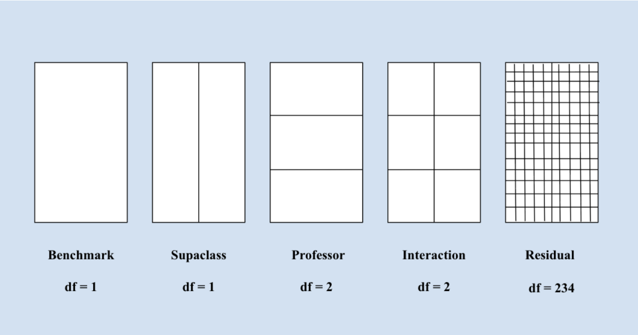

Note that the residual component does not actually have to have 234 squares drawn out; stating the degrees of freedom will suffice.

Problem 1 - Solutions for Part B and C

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The effect of Supaclass on the final exam scores is dependent on which professor the student is taking STATS 101C with.

Problem 2

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A.) For the first experiment, how large should the sample size be to detect a difference of 10 points or more?

B.) For the second experiment, how large should the sample size be for comparing the two group means to detect a difference between means of 5 points or more?

Problem 2 - Solutions

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$$n = \left[\frac{(z_{1-\alpha/2} + z_{1-\beta})\sigma}{\text{diff}}\right]^2 \approx \left[\frac{(1.96 + 0.842)20}{10}\right]^2$$

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B.) For the second experiment, how large should the total sample size be for comparing the two group means to detect a difference between means of 5 points or more?

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B.) For the second experiment, how large should the total sample size be for comparing the two group means to detect a difference between means of 5 points or more?

$$n = \left[\frac{2(z_{1-\alpha/2} + z_{1-\beta})\sigma}{\text{diff}}\right]^2 \approx \left[\frac{2(1.96 + 0.842)20}{5}\right]^2$$

$$n \approx 502.48 \implies n \geq 503$$

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- **Supaclass**: True, False
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Hence, there will be ten students in each treatment combination.

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After collecting the data, he decides to run two-way ANOVA, with the output coefficients being the following:

Problem 3 - Summary Output

```
Coefficients:
                                Estimate Std. Error t value Pr(>|t|)
(Intercept)                       80.800      3.020  26.751  < 2e-16 ***
supaclassTRUE                      0.300      4.271   0.070  0.94427
exerciseCardio                   -10.300      4.271  -2.411  0.01933 *
exerciseWeighttraining             4.000      4.271   0.936  0.35321
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D.) Given that the overall mean of the final exam scores is 79.68, calculate the main effect values for each level of the three levels of exercise type.

Problem 3 - Solutions to Part A and B

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It represents the average final exam score among students underwent calisthenics and did not receive Supaclass grading.

B.) Given that the standard deviation of the final exam scores is about 11.11, calculate the total sum of squares.

$$n = 60 \implies \text{df}_{\text{total}} = 59$$

$$SST = \sigma^2 \times \text{df}_{\text{total}} = 11.11^2 \times 59 \approx 7282.49$$

Problem 3 - Solution to Part C

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$$R^2 = 1 - \frac{SSE}{SST}$$

$$0.323 = 1 - \frac{SSE}{7282.49} \implies SSE \approx 4930.25$$

Problem 3 - Solution to Part D - I

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$$\tau_{calisthenics} = \bar{x}_{calisthenics} - \bar{x}_{overall} = 80.95 - 79.68 = \mathbf{1.27}$$

Problem 3 - Solution to Part D - II

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$$\bar{x}_{cardio} = \frac{10 \cdot (80.8 - 10.3) + 10 \cdot (80.8 + 0.3 - 10.3 + 17.4)}{20} = 79.35$$

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Problem 3 - Solution to Part D - III

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$$\bar{x}_{weights} = \frac{10 \cdot (80.8 + 4.0) + 10 \cdot (80.8 + 0.3 + 4.0 - 12.4)}{20} = 78.75$$

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$$\tau_{weights} = \bar{x}_{weights} - \bar{x}_{overall} = 78.75 - 79.68 = \textbf{-0.93}$$

Problem 3 - Part D - General Framework

Suppose that Factor A has k levels, indexed by $i = 1, ..., k$, and that Factor B has m levels, indexed by $j = 1, ..., m$. Assume that each treatment combination contains n observations.

For the group-level means of Factor A,

$$\bar{x}_i = \frac{n(\beta_0 + \beta_i) + \sum_{j=2}^{m} n(\beta_0 + \beta_i + \beta_j + \beta_{i\times j})}{mn}$$

And for the group-level means of Factor B,

$$\bar{x}_j = \frac{n(\beta_0 + \beta_j) + \sum_{i=2}^{k} n(\beta_0 + \beta_j + \beta_i + \beta_{i\times j})}{kn}$$

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